

SynDiff Synthetic Benchmark — Live Progress

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13/942 jobs complete (1.4%)

$S = I(X_1; Y|X_2) - I(X_1; Y)$ in nats; $S > 0$ synergy-dominant, $S < 0$ redundancy-dominant. SynDiff (ours) vs prior estimators.

Sweep progress

Phase	Method	Done	Expected
P1	batch_pid	0	96
P1	broja	0	6
P1	infonce	3	96
P1	minde_comp	0	96
P1	mine	3	96
P1	plugin	2	6
P1	syndiff	5	96
P2	batch_pid	0	90
P2	infonce	0	90
P2	minde_comp	0	90
P2	mine	0	90
P2	syndiff	0	90

Accuracy vs ground truth (completed cells, lower error better)

Method	#cells	mean $ \text{est} - \text{truth} $	sign acc
syndiff	2	0.0435	1.00
plugin	2	0.1343	1.00
mine	1	1.0078	1.00
infonce	1	6.9437	0.00

SynDiff vs prior — head to head (nats)

Cell	truth	SynDiff	batch_pid	minde_comp	mine	infonce	plugin
p1_dial_rhon0p9_sy0p1_d8_n20k	+1.473	+1.412	–	–	+0.466	-5.470	–
p1_dial_rhon0p9_sy1_d8_n20k	+0.083	+0.058	–	–	–	–	–